STARTUP ZONE

Saving Lives Through ECG At Home

CardioMon ECG monitor and analyser

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Myocardial infarction, or in simple terms heart attack, is one of the major ailments that lead to human fatality, especially in a country like India. This is primarily due to a sedentary lifestyle and diet. It is crucial to identify a cardiovascular disease and immediately rush the patient to the hospital. However, the main challenge lies in recognising the symptom(s) of the ailment.

Myocardial infarction has indirect symptoms like back pain, shoulder or arm ache, acidity and so on. Most people cannot relate these symptoms with the disease. Ideally, the person suffering from these symptoms should be rushed to the hospital within an hour—called 'golden hour'—to reverse the effects of the condition. But unawareness and lack of a portable electrocardiography (ECG) or analysis device to substantiate the condition causes concerning delay in action.

Moreover, most general physicians do not have an ECG machine in their clinics or carry it during home visits. This is due to the high cost and form factor of existing, conventional bulky ECG machines. As a result, they send patients to a hospital to get an ECG test done. Critical time is lost here, without any preliminary medication administered to the patient.

To fight this shortcoming, Anuj Sharma, founder and lead scientist of Anusandhan Technologies, designed and developed a portable ECG monitor and analyser called CardioMon.

Sharma recalls, "One of my relatives was experiencing symptoms of a heart attack when he was about to board a train. Since there was no device to detect it, he had to be rushed to the hospital. Experiencing this problem first hand, I decided to use my expertise in electronics and embedded systems to see if I could find a solution."

Building CardioMon: the working principle

Sharma took this challenge as a personal research project initially, using digital signal processing (DSP) techniques to detect myocardial infarction. He says, "After four years of research, iterative design and development, we gradually got success on our prototype. I redefined the design, ensured accuracy and shaped it into a complete product, called CardioMon." With CardioMon, a real-time echocardiogram can be displayed on the screen. Since it is Bluetooth-enabled, results can be recorded on a portable 7.6cm (3-inch) Bluetooth printer.

Simply put, CardioMon is a miniaturised, hand-held, battery-operated, 12-lead ECG monitoring and analysis device, which enables patients and doctors to take immediate and informed medical decisions during cardiac emergencies.

CardioMon device (ECG monitor and analyser) has an inbuilt 320x240 pixel TFT colour LCD display to view results. Input is drawn through 10 electrodes: four clamp electrodes to be connected to four limbs and six bulb electrodes to be placed on the chest at specific positions.

An ECG cable is used to connect the electrodes to CardioMon device. Electrodes can be placed directly on the body, or be used with ECG gel to improve contact. Like any standard 12-lead ECG device, it can provide 12 ECG waveforms. Based on these signals, the devised algorithm provides a preliminary indication in case of myocardial infarction.

Further, Sharma has also developed an exclusive vest for the device to allow end consumers to take accurate ECGs easily. He says, "Electrodes have to be placed in specific positions to get the right results. While that is easy for doctors and paramedics, patients are mostly unaware. To address this, we have designed a vest that, once configured, can position the electrodes by itself."

The device can be purchased by medical institutions, doctors, paramedics as well as end consumers. The